>> Intelligent Systems – WC 5<<

Vagueness, Probabilities and Bayesian Networks

Concepts to be known after WC5 :

- Uncertainty versus Vagueness
- Fuzzy Logic, Fuzzy Relations
- Probabilistic Knowledge
    - Independent/dependent events (+ calculation rules)
    - Probability
        - Joint, prior ('unconditional'), posterior ('conditional')
    - Bayes' theorem
- Bayesian Networks

Exercises Vagueness, Fuzzy Logic and Fuzzy Relations

Exercise 1: Consider the following statement: "Half of the Dutch men are married". Which of the following statements is true?

A) The author of this statement describes the marital status of Dutch men with a degree of vagueness.
B) The author of this statement describes the marital status of Dutch men with a degree of uncertainty.

Give an example for the "other" statement?

Exercise 2: The director of our institute wants to reduce the environmental impact of the company by reducing the CO2 emissions caused by transportation. Therefore, she wants to give a devise a policy to support those who do not live very close (they can come by bicycle) or those who live very far (they should do remote work).

Since you know the director is a big fan of fuzzy set theory, you decide to model this problem with fuzzy sets. How do you model the fuzzy set of "employees who live moderately far away", i.e. those who do not live close, but also not far away?

Exercise 3: Given the following (standard) semantics for Fuzzy Logic for the following operators:

- f(¬A) = 1 - f(A)
- f(A v B) = max(f(A);f(B))
- f(A & B) = min(f(A);f(B))
- f(A -> B) = min(1;1 – f(A) + f(B))

In Fuzzy logic the meaning of a formula is defined as a value between 0 and 1. What is the meaning of f(((A -> B) v B) & (¬A v ¬B)) if f(A)=0.8 and f(B)=0.5?

Exercise 4: Suppose that there are two relations defined as follows, where L5 and L6 stand for Lecture 5 and 6 respectively:

understands

	L5	L6
John	0.4	0.6
Peter	0.3	0.7
Julia	0.1	0.4

contains

	semantics	reasoning
L5	0.8	0.3
L6	0.4	0.8

Fill in the table for the composition of the two relations, i.e. the relation

understandslecturecontaining = understands o contains

	semantics	reasoning
John		
Peter		
Julia		

Exercises probabilistic knowledge

Exercise 5: Consider the full joint probability distribution P(A,B,C) shown in the table to the right:
a) What is P(A = true)?
b) What is P(B ^ C = true)?

A	B	C	P(A, B, C)
false	false	false	0.05
false	false	true	0.10
false	true	false	0.03
false	true	true	0.25
true	false	false	0.15
true	false	true	0.02
true	true	false	0.07
true	true	true	0.33

Exercise 6: Given the full joint distribution as above, calculate the following:

	toothache		*¬ toothache*	
	catch	*¬ catch*	*catch*	*¬ catch*
cavity	.108	.012	.072	.008
¬ cavity	.016	.064	.144	.576

a. P(toothache).
b. **P**(Cavity).
c. **P**(Toothache|cavity).
d. **P**(Cavity|(toothache v catch)).

Exercise 7:
You throw two fair dice, one green and one red, and observe the numbers uppermost. Decide whether the following pairs of events are (in)dependent.
A: the sum is even
B: the red die is even

Exercise 8:
You throw two fair dice, one green and one red, and observe the numbers uppermost. Decide whether the following pair of events are (in)dependent.
A: the sum is 5
B: the red die shows a 2

Exercise 9:
Given the following information:

- The probability that someone with a cold also has fever is 80%.
- The prior probability of having a cold is 0.0001.
- More or less 1 out of 1000 people in the population has fever.

I have fever. Calculate with Bayes' Rule the probability that I have cold.

Exercise 10: Bayesian Networks

Given the following Bayesian network, what is the

- Probability that an earthquake happens?
- probability that the alarm rings?
- Probability that john calls?

B	P(B)
+b	0.001
-b	0.999

E	P(E)
+e	0.002
-e	0.998

B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

A	J	P(J|A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99